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Interactive exploration of benchmarking results with *bettr*

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Software

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Summary

Method benchmarking is a core part of many research fields and aims to establish best practices in method selection and application as well as help identifying gaps and possibilities for improvement in existing methods. A typical benchmarking study scores a set of methods using a variety of different metrics, intended to capture different aspects of performance and usability. For practical purposes, e.g., when a user needs to select a method for their own analysis, it is often the case that not all of the evaluated metrics are equally important. For example, the ability of a statistical method to handle arbitrarily complex experimental designs may not be relevant if the user only needs to perform a two-group comparison, while memory frugality may be of high importance if the analysis will be performed on a system with limited computational resources. Hence, having the ability to easily and adaptively create a weighted summary score by which the methods can be ranked, or alternatively to create a visual summary of only the desired metrics, would make it possible for users to tailor their method choice based on the aspects that are most relevant to them.

Inspired by the OECD 'Better Life Index', which allows users to score OECD countries by a customizable weighted average of various factors contributing to human well-being, we developed the *bettr* R/Bioconductor package to provide similar support for consumers of benchmarking studies. Given a table with values of evaluation metrics across the evaluated methods, as well as optional method annotations and metric groupings, *bettr* allows users to visualize performance summaries emphasizing the aspects and evaluation metrics that are most important to them (Figure 1). Examples of metrics include the area under the receiver operating characteristic (ROC) curve for a classification task, or the execution time. Method annotations (e.g., whether they are designed to work on untransformed or log-transformed input data) allows the user to evaluate whether methods with certain characteristics tend to perform better than others. *bettr* can be used interactively as an R/Shiny application (Chang et al., 2025), or programmatically by calling the underlying functions directly for full reproducibility and integration into analytical pipelines (see the package vignettes for more details).

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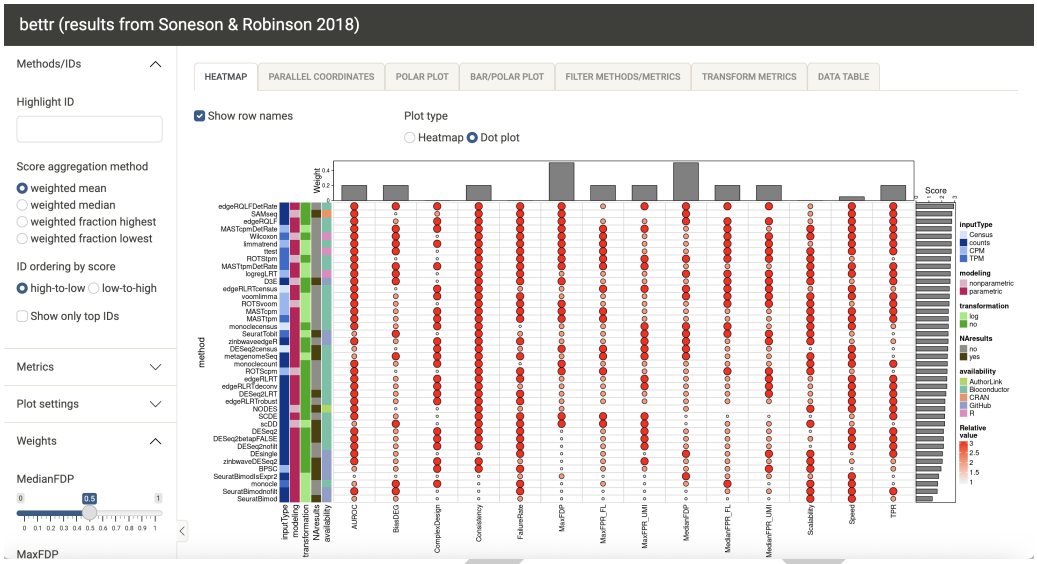

Figure 1: Screenshot of a *bettr* application illustrating the benchmarking results from Soneson & Robinson (2018). The size and color of the circles indicate the evaluation score for each method (rows) and metric (columns), with higher values being more favorable. The bars above the plot indicate the current weights for the metrics (these are controlled by the sliders in the bottom of the left sidebar), and the bars to the right of the plot represent the aggregated scores for the methods (in this case, the weighted mean across metrics). The colored bars to the left of the plot represent categorical annotations and characterizations of the methods, which are not taken into account when determining the ranking. In addition to the displayed dot plot, the results can be visualized in several other ways, including a heatmap, polar plots, and using parallel coordinates. Moreover, the set of methods and metrics to include in the display can be controlled by the user via the 'Filter methods/metrics' tab.

bettr accepts input in multiple formats, including a collection of data frames, a SummarizedExperiment object (Morgan et al., 2025) or a JSON file, and contains functions for converting one input format to another. This increases the flexibility of the application and makes it easy to combine with a variety of workflows. For example, a user working locally in R may find it more convenient to generate a set of data frames, while an automated benchmarking workflow could export all necessary components in a single, platform-independent JSON file. *bettr* can also be deployed in server mode, allowing users to upload their own input data (in JSON format) to a running app.

Statement of need

The interactive nature of *bettr* makes it particularly suitable for visual, exploratory analysis of benchmarking results, since the user can interactively modify the weights associated with the different evaluation metrics and immediately see how it affects the ranking of the methods. It is also straightforward for authors of benchmarking studies to deploy an instance of *bettr* using, e.g., a local Shiny server or a commercial option such as https://shinyapps.io, to allow readers to easily explore their results. Complementary functionality for creating static summary representations (e.g., heatmap-like visualizations) of benchmarking results is provided e.g. by the funkyheatmap package (Cannoodt et al., 2025), and benchmarking platforms such as OpenEBench (Capella-Gutierrez et al., 2017) and OpenProblems (Luecken et al., 2025) also produce result visualizations for the included benchmarks. However, existing tools typically lack either flexibility in input format or means of deployment, or the ability to explore results interactively, and are not intended to support user-specific metric weighting.

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Availability and examples

bettr is available via Bioconductor and on GitHub. A collection of example data sets and *bettr* configurations are available from https://github.com/csoneson/bettr-examples. In addition, an example instance, using data from Soneson & Robinson (2018), is deployed on https://csoneson.shinyapps.io/soneson2018de/.

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